

# Testing the Validity of CoIG's Causal Relevance Metric

## *Instructions*

Proposals must be specific and detailed so the direction of the research is clear. If you're not sure about a given item, please mark it as such and we can discuss.

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## Relevant Past Papers

*How is it done today, and what are the limits of current practice?*

- Title: Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting
  - Link: <https://arxiv.org/pdf/2305.04388>
  - Summary: This research shows that the logical reasoning of language models could be realistic yet misleading about the way the model makes decisions. Hidden biases introduced to prompts make the model provide an extremely real justification of its result without ever referring to the true reason.
  - Relevance: This shows the issue our proposal will address in the field of imagery. The assumption that step-by-step thinking must always be faithful simply because it appears coherent to us.
- Title: Sanity Checks for Saliency Maps
  - Link: <https://arxiv.org/pdf/1810.03292>
  - Summary: In this paper, it was shown that many interpretability heat maps that had been in use for quite some time were basically just edge detection models and not the explanation metrics that researchers assumed. This was proven through randomly initializing the weights in the model, allowing us to check the sanity of the models.
  - Relevance: This can be seen as the conceptual foundation of our proposal. In the same way that the paper established that saliency maps were totally confused by visual elements, the proposed experiment intends to test whether Causal Relevance is also confounded by the CoIG compositional lock.
- Title: Prompt-to-Prompt Image Editing with Cross Attention Control

- Link: <https://arxiv.org/pdf/2208.01626>
- Summary: The layout and geometry of the AI-generated images are defined by the cross-attention maps in the initial stages of the diffusion process. Through the manipulation of these early-stage attention maps, keeping the layout is achieved despite the changes in text prompts.
- Relevance: This shows the actual mechanics behind why CoIG needs a compositional lock at all. Knowing how diffusion models actually lock down the maps very early in the generation will be important for interpreting the findings if the metric is found to be wrong.
  
- Title: Measuring Faithfulness in Chain-of-Thought Reasoning
  - Link: <https://arxiv.org/pdf/2307.13702>
  - Summary: Causal faithfulness of the LLM reasoning chain is evaluated using the approach of intervention of the intermediate steps, in many different ways, including truncation, error injection, and modification. The authors find that the level of influence of the intermediate steps on the output is very different between models and tasks.
  - Relevance: This paper has been sort of an inspiration for the experimental design proposed by Chayan. Intervention on intermediate stages (which we are performing by using "shuffled" and "irrelevant" image sequences) is described in the paper as the established practice for assessing whether the model relies on its prior stages, which therefore allows us to test the faithfulness of the CoIG model.
  
- Title: Faithful Chain Of Thought Reasoning
  - Link: <https://arxiv.org/pdf/2301.13379>
  - Summary: Here, they propose a framework which establishes faithful by construction reasoning by splitting the model into a translational (converting query into a symbolic chain) and problem solving stage (producing answer). Now the output is created by executing reasoning instead of generating reasoning alongside it without any accuracy lost.
  - Relevance: This paper doesn't test whether steps cause the output, it ensures dependence, so we can make clear comparisons to real step-by-step faithfulness to CoIG. Additionally, it establishes that the generated chain may not semantically reflect the intended image (since it wasn't studied in the paper), which is our gap.
  
- Title: Can We Generate Images with CoT? Let's Verify and Reinforce Image Generation Step by Step
  - Link: <https://arxiv.org/abs/2501.13926>
  - Summary: The paper introduces a framework called PARM, which adds step by step verification to image generation. It trains a separate reward model that actively checks the image in intermediate steps. If a step does not accurately

match the prompt, the model flags the error and chooses a better path of generation.

- Relevance: This is just a good point of comparison for our project. It demonstrates that the field is heading towards dynamic AI verifiers rather than hardcoded locks. We can use this paper to argue why auditing CoIG's rigid lock mechanism is necessary to see if older frameworks are reliable.
- Title: What the DAAM: Interpreting Stable Diffusion Using Cross Attention
  - **Link:** <https://arxiv.org/abs/2210.04885>
  - Summary: This paper introduces DAAM (Diffusion Attentive Attribution Maps), a method for visualizing how different words in a text prompt influence specific regions of an image generated by Stable Diffusion. The authors show that cross attention maps capture meaningful relationships between text inputs and image generation, allowing researchers to interpret how diffusion models process and respond to prompts.
  - Relevance: This paper provides background on interpreting diffusion models through their internal attention mechanisms. It supports our project by demonstrating that intermediate processes within image generation models can be analyzed, which is important when evaluating whether CoIG's intermediate generation steps are genuinely contributing to the final output.
- Title: High-Resolution Image Synthesis with Latent Diffusion Models
  - **Link:** <https://arxiv.org/abs/2112.10752>
  - Summary: This paper introduces Latent Diffusion Models, which perform the diffusion process in a compressed latent space rather than directly on pixels. The authors demonstrate that this approach improves computational efficiency while maintaining high quality image generation, making it the foundation for modern text to image systems such as Stable Diffusion.
  - Relevance: This paper provides essential background for our project because CoIG is built on diffusion based image generation. Understanding how diffusion models progressively construct images helps explain why intermediate generation steps exist and why mechanisms such as the compositional lock are important when studying whether these steps have a genuine causal influence on the final output.

Title: Probing and Steering Chain-of-Thought Unfaithfulness in Language Models

Authors: Giovanni Maria Occhipinti, Alessandro Abate, Nandi Schoots

Venue: ICLR (TTU Workshop, Main Track Oral)

Year: 2026

Link: <https://openreview.net/pdf?id=JL8sNbnSWK>

Summary: Investigates chain-of-thought faithfulness by probing a model's internal (latent)

representations rather than relying only on its text output. The authors identify an "honesty" direction in the middle-to-late layers that encodes whether a model's stated reasoning matches its internal processing, and show that linear steering along this direction can make generated reasoning up to 46% more faithful.

Relevance: This establishes that step-by-step output can appear well-structured and coherent while concealing unfaithful rationalization internally. It motivates why ColG's mechanics must be tested directly: if language models require white-box steering to stay honest, we cannot assume an image model's external metric is valid without testing it against controlled failures.

Title: How Does Unfaithful Reasoning Emerge from Autoregressive Training? A Study of Synthetic Experiments

Authors: Fuxin Wang, Amr Alazali, Yiqiao Zhong

Venue: arXiv (preprint, cs.LG)

Year: 2026

Link: <https://arxiv.org/abs/2602.01017>

Summary: Uses synthetic experiments to study why unfaithful reasoning arises during autoregressive training. The authors find that under noisy data or increased task difficulty, models undergo a sharp transition away from executing genuine step-by-step computation and toward producing reasoning chains as a surface-level format, while internally relying on a shortcut to reach the answer.

Relevance: This gives direct backing to our hypothesis by showing that models tend to generate faithful-looking but non-causal reasoning under pressure. It is precisely the failure mode we test for: whether ColG produces a plausible-looking chain of steps while the compositional lock, rather than the steps' semantic content, actually determines the final image.

## Motivation

- Things to include:
  - What limitation or problem are you solving and how do you know it exists
    - ColG introduces Casual Relevance to prove that steps are faithful but the metric doesn't support that since the compositional lock freezes each step's content to later steps and doesn't overwrite it. So the step persists to the end, so either faithfulness or the lock drove the Casual Relevance metric. This is ambiguous because the ColG paper only tested real/semantically coherent chains of thought, so it cannot be inferred which of the two explanations drove the Casual Relevance.
  - Why is this limitation important
    - Causal Relevance is the one piece of evidence ColG uses to go from steps being readable to steps being faithful, and that jump is the whole reason ColG can be monitorable. Monitorability matters for safety because it's what lets us trust the system, and it depends on reasoning being causally faithful. This is a problem with the measuring tool itself,

which comes before any question about whether CoIG's images look good.

- Why does your idea solve it
  - The whole issue is that faithfulness and the lock are tangled together in every chain CoIG tested, so the fix is to pull them apart by building chains where they predict opposite results. We deliberately make unfaithful chains, where the sub-prompt's meaning does not match the content generated at that step, giving us a case with known ground truth. If Causal Relevance still scores high, the metric is measuring the lock, since meaning is the only thing we removed; if it drops, the metric is tracking real semantic influence like it's supposed to. Either way the ambiguity is resolved. This is the same approach Zaman and Srivastava used in text to show the "Biasing Features" metric confuses real unfaithfulness with the lossy compression of natural-language explanation — we're porting it to images, where it hasn't been done.
- Why would your idea probably work
  - First, the confound only exists on coherent chains, so adding incoherent ones is a move that's guaranteed to separate faithfulness from the lock. Second, the strategy already worked in the text domain, where controlled ground-truth cases exposed hidden conflation in an established metric, so it has precedent. Third, the failure mode we're testing has no analog in text: in LLMs a step exerts influence through the model's own computation (which is what tests like Filler Tokens measure), but CoIG bolts on an external mechanism that guarantees persistence regardless of whether the sub-prompt drove the content.

## Key Ideas/Contributions/Novelty

- What are we doing that hasn't been done before? Things to consider:
  - What is the novelty of the idea with respect to the existing literature? What is "new"?
  - What is the main contribution (to science) of this paper? Are we creating an improved method or a benchmark? What new insights or findings are we providing? What research question are we answering that was not answered?

Novelty: We added negative-control testing to T2I generation, which is usually the best way to test for causal faithfulness in LLMs. We also found that CoIG's compositional lock forces visual persistence, so we based our hypothesis on the idea that it inflates the metric, which identifies a unique issue that no one else has addressed. We also isolated that CoIG's original eval. Method only tested coherent chains, but our design holds the lock while breaking the prompt order and semantics, thereby separating the mechanical lock from faithfulness and allowing us to see whether they correspond, which is the basis of our experiment.

Currently, image benchmarks evaluate image quality or text alignment, but we are creating an evaluation set of broken visual reasoning chains, allowing us to create a rare, objective ground truth for unfaithfulness in generative images.

Contributions: We are moving beyond merely auditing an existing interpretability tool. Our primary contribution is the development of a novel, architecture-agnostic visual faithfulness metric specifically designed for chain-of-image generation. By translating "Thought Anchor" methodologies from LLM reasoning into the visual domain, we provide a robust evaluation standard for future dynamic AI verifiers (like PARM) that separates true causal influence from hardcoded state persistence.

New Findings:

1. A definitive hardware-level check on whether CoIG's compositional lock acts as a mechanical confound in current generation evaluation.
2. The establishment of a new evaluation metric that measures the actual causal weight of intermediate visual steps, proving which latent reasoning states matter for the final output regardless of the prompt's textual coherence.

Research Question: "Does CoIG's Causal Relevance metric measure genuine, step-level semantic faithfulness, or does it merely detect the mechanical enforcement of its own compositional lock?" By answering this question, we can safely know whether the CoIG is actually reasoning instead of just creating the image mechanically and lying to the human.

Moreover, one of the important contributions of this study is the formulation of a replicable methodology template for the generative vision research community. As more text-to-image models employ step-wise or dynamic verification approaches, there will be an increased demand for standardized auditing tools. Through the creation of the procedure of designing, conducting, and analyzing a matched negative control experiment within the visual generation pipeline, we provide a generalizable methodology template. Future studies can utilize our experimental design with three conditions (Real, Shuffled, and Substituted) to audit the measures of completely different visual generation models.

Also, this study presents an innovative theory connecting NLP safety with computer vision interpretability. For many years, the safety researchers working on LLMs have used negative control tests like corrupting intermediate processes or using filler tokens in order to demonstrate if the machine is doing real reasoning or simply "hallucination" of a logic process. But this approach has not been firmly applied to image generation. This experiment represents the first attempt to apply the rigorous NLP stress testing approach to the visual domain and measure "unfaithful reasoning" in a pipeline where physical constraints apply.

In addition to the above, even though this is not a complete novelty in terms of designing a new benchmark from scratch, our experimentation itself produces a unique and extremely valuable test set: a carefully constructed dataset of intentionally "broken" visual reasoning chains. With a

fixed architectural lock and the intentional corruption of the semantics provided by the prompts, we produce hundreds of examples of unfaithful visual generation. This particular dataset of Shuffled and Substituted image chains will be our first-ever ground truth for unfaithful visuals that will help the entire community of AI safety experts understand what happens when the model's inner logic is disconnected from its outputs.

## Methods

- How does your idea work? Describe the way you will get your results from the initial step. Make a diagram.
  - This is mainly just a matched negative-control experiment where we take the ColG generation code from the original paper authors and generate the per-step images only once per prompt — under the real, correctly-ordered sub-prompts. The Shuffled and Substituted conditions do not trigger new image generation; they reuse that same fixed set of per-step images and only change which sub-prompt text is presented as the "instruction" for each step. The compositional lock stays switched on throughout, so if Causal Relevance is actually measuring faithfulness to the (now mismatched) sub-prompt, the score must drop in the Shuffled/Substituted conditions. However, if the score remains the same, it was only ever measuring the lock's mechanical persistence of the original content.

Given the possible confound due to the compositional lock, our methodology will be divided into two parallel streams:

### Track 1: Check on Lock Mechanism

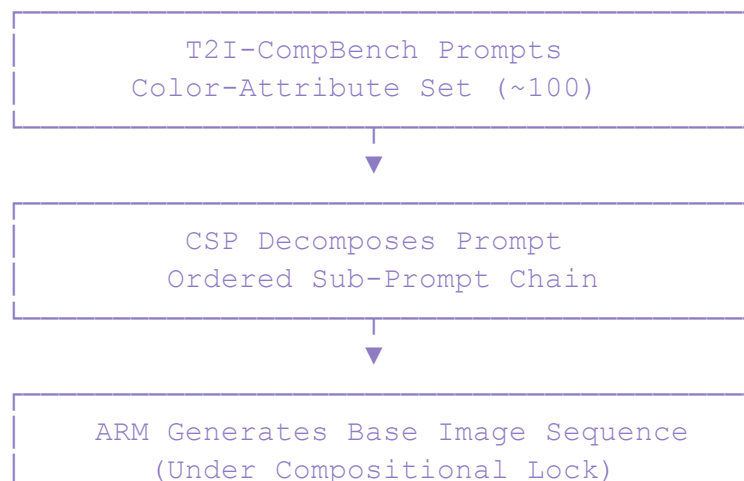
Before conducting a full audit using metrics, we conduct a diagnostic check of the lock mechanism. We subject a few prompts to the ARM system without using the compositional lock, followed by using it. The baseline pixel drift from frame to frame will determine whether there is a mechanical persistence forced regardless of any textual input.

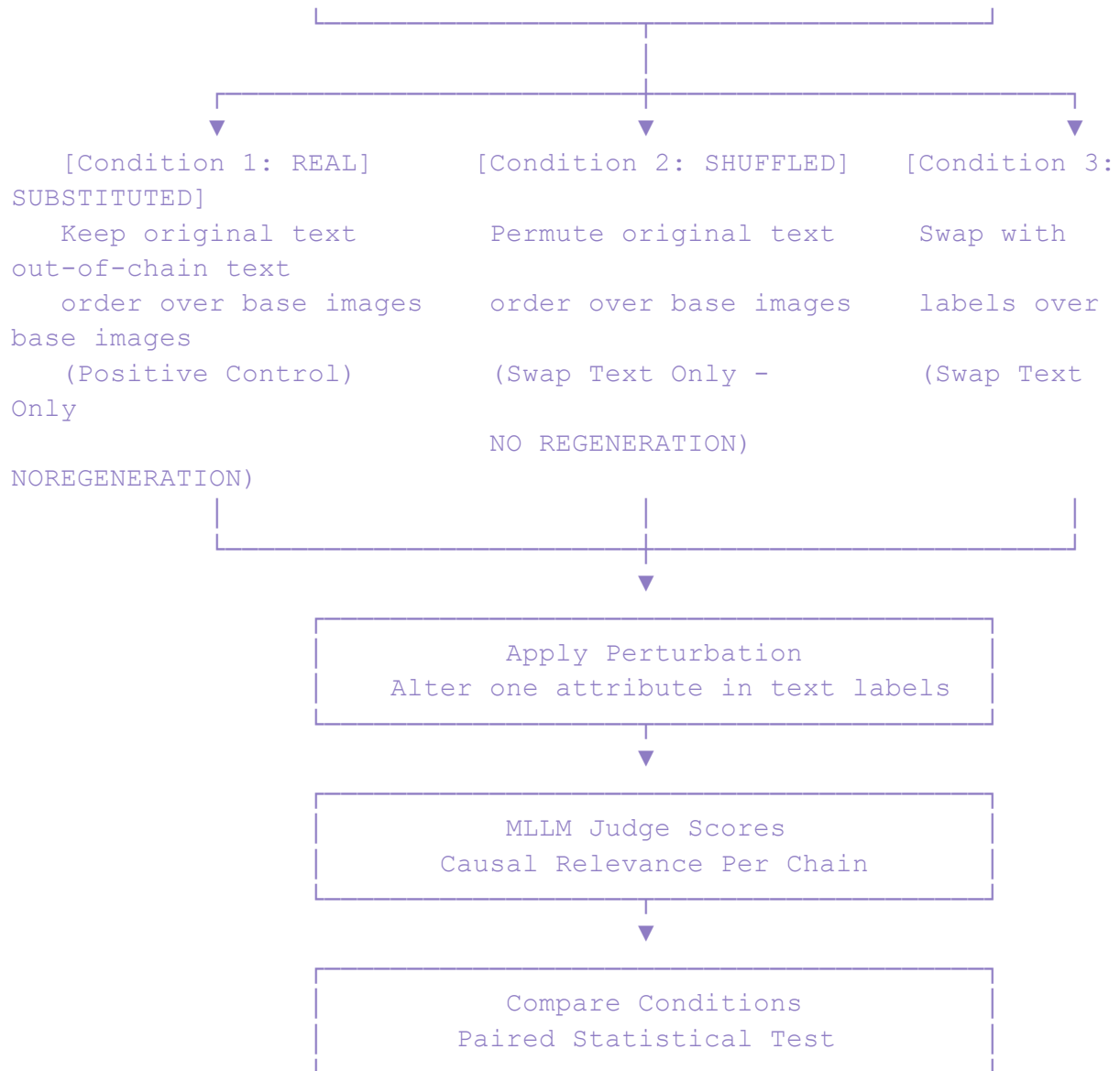
### Track 2: Constructing a New Visual Faithfulness Metric (The Anchor Metric)

Instead of only utilizing the Causal Relevance of ColG, we construct a metric inspired by "Thought Anchors." In this process, we use interventions such as noise injections and/or layer truncations on the latent spaces in the intermediate stages of image generation. Using structural similarity (SSIM) and CLIP semantic drift between the intervened and un-intervened images, we can compute a "Causal Weight Score" at each stage.

- 1) From T21-CompBench (same subset ColG paper uses) we take around 100 color-attribute prompts
- 2) We use the compositional strategy planner from the original authors to decompose the prompt into a specific sequence of sub-prompts

- 3) We generate the per-step images exactly once per prompt, using the authors' Autoregressive Refinement Model (ARM) under the compositional lock, driven by the real, correctly-ordered sub-prompts. This one fixed image sequence is reused across all three conditions — no new images are ever generated from the broken sub-prompts.
- 4) Most importantly, we do not regenerate any intermediate images at each step. We create two incomplete chains by only exchanging the sub-prompt text overlaid onto the previously generated image sequence at step 3.
  - Shuffled (Testing Sensitive to Order): Original sub-prompts are assigned to fixed images in a randomly permuted order. Here, the Causal Relevance metric tests whether it is only sensitive to the order of steps, since all sub-prompts still have something to do with the resulting image globally but are out of sync with its generation.
  - Substituted (Testing Fully Unrelated Texts): At each step, the text label is entirely swapped for some other sub-prompt from another image chain sequence. Here, we test our metric against the ground truth of unfaithfulness; there is no semantic relation whatsoever between the text and the generated image at this step.
- 5) Then we use the perturbation protocol and for every chain alter 1 attribute in the sub-prompt text assigned to a step, like changing "red bowl" to "blue bowl" — without regenerating the image
- 6) Then we judge by using the authors' MLLM judging pipeline. For every chain, we check if the altered attribute both appears and persists to the final image — even though, for Shuffled and Substituted, that image was never actually generated from the altered text. This leads to getting a Causal Relevance score for each chain.
- 7) Finally we interpret the Mean Causal Relevance across the three conditions using a paired test. The results (either  $\text{Real} \approx \text{Shuffled} \approx \text{Substituted}$  /  $\text{Real} > \text{Substituted}$  only /  $\text{Real} > \text{Shuffled} > \text{Substituted}$ ) shows whether the metric is fully confounded, partially sensitive, or genuine.





## Experimental Setup

- How are you going to test your idea to prove that it works?
  - Think about social science experiments where researchers have a plan of what to make participants do, what to ask them and how to calculate results based on the responses.
  - What are your baselines for comparison? i.e control group
  - What models? What datasets? What metrics (if not accuracy)?

Most importantly, across those same images, we'll compare the three textual treatments to the label associated with those sub-prompts (real, shuffled and substituted), so that none of our sub-prompt texts will be novel and generate images for the negative controls. We first obtain the original text prompts from our dataset. We then use a Compositional Strategy Planner to generate an ordered list of text sub-prompts corresponding to each step.

We pass the sequence through our autoregressive refinement model once with the compositional lock on, generating per-step images for this valid, properly ordered sequence, which acts as our baseline.

Importantly, we will use three textual treatments to the labels for these sub-prompts over that same set of images, ensuring no sub-prompt texts will lead to new images in our negative controls. Lastly, we will inject an attribute token perturbation into the sub-prompt text at one intermediate step in each sequence and will then ask an MLLM judge if the attribute token appears in the (unchanged) final generated images, providing a metric score.

Baseline for Comparison: We will compare the results to groups of automatically generated prompts: A positive control group (real) - the true text sequence for our ordered, fixed set of images. This represents the true metric value under normal operation.

Two negative control groups will compare with identical fixed image sets but with altered sub-prompt texts.

- Shuffled replaces sub-prompts with those from other real chains (not another generation process entirely), tested to ensure that the system does not learn based purely on ordering of sub-prompts.
- Substituted replaces sub-prompts with random text, guaranteed to not match the contents of generated image for that step, testing ability of metric to detect lack of semantic connection to image at a particular step.

Models: We use the exact open-source implementations from the ColG repository to avoid implementation discrepancies: Compositional Strategy Planner (CSP), Autoregressive Refinement Model (ARM), MLLM judge pipeline

Datasets: We use the T21-CompBench dataset, isolating approx 100 color-attribute prompts. This directly mirrors the specific evaluation subset used in the original ColG publication to ensure consistency.

Metrics: The primary dependent variable is ColG's casual relevance score rather than accuracy.

- What additional analysis do you plan on doing?
  - How are you going to prove that any improvements in the system are because of your idea alone, i.e. reduce confounding factors and do ablation testing
  - How does your idea impact the system? What metrics are you quantifying with?

- Visualizations? Statistics?

Aside from generating averages, we plan to map out a complete sensitivity profile of the metric by evaluating its performance across varying prompt structures and testing boundaries . This allows us to see whether the metric responds uniformly to text modifications or if it shows isolated vulnerabilities under specific conditions.

To isolate the variable, semantic faithfulness, from architectural confounders, we implement an explicit architectural control setting. We reduce conflation using a Constant Architecture; the underlying model architecture, prompt lengths, neural weights, and importantly, the compositional lock will be kept completely unchanged and on across all conditions. Since meaning is the singular varied item between the real, shuffled, and substituted groups, any significant discrepancies in the scoring outputs can be attributed to semantic tracking rather than the mechanical persistence of the lock layout.

How our idea impacts the system, and what metrics we are quantifying with: The framework functions as an external safety auditing pipeline, it does not structurally alter or retrain the generative system. We quantify metric reliability by analyzing the mathematical relationships between the resulting scores:

$\text{ScoreReal} \approx \text{ScoreShuffled} \approx \text{ScoreSubstituted}$  proves that the metric is fully confounded by the lock and fails to actually measure true casual faithfulness.

$\text{ScoreReal} > \text{ScoreSubstituted}$  and  $\text{ScoreReal} \approx \text{ScoreShuffled}$  maps partial coarse-grained sensitivity.

$\text{ScoreReal} > \text{ScoreShuffled} > \text{ScoreSubstituted}$  validates the metric as a granular reliable tracker of semantic faithfulness.

Visualizations: Performance will be visualized through cross-condition comparison plots detailing score distributions across the evaluation cohort, combined with a structural method diagram that illustrates the negative-control pipeline layout.

Statistics: We will compute the mean values and standard deviations for the casual relevance scores across all three test arms. We will apply statistical significance testing to definitively classify the metric's behavior into one of our three hypothesized profiles (see directly above).

## Datasets and Evaluation

- Which datasets are you going to use to evaluate your method? Or, are you creating your own?
  - T2I-CompBench's color binding subset serves as a source of prompts to be used in our experiments, so it is the very benchmark that CoIG employs for its Causal Relevance evaluation. This way, our results can be directly compared to those of CoIG. We need to provide an evaluation mechanism for each prompt in

T2I-CompBench. We produce only one real image sequence and then create two negative control sequences by relabeling the sub-prompt texts of the image sequence without generating any additional images.

- If you're training, what dataset will you use for that?
  - We don't train anything as every model is from the authors github repos: CSP, ARM, and the MLLM judge are from them. The only data that is produced is from the generated chains and their casual relevance scores which we obtain later on.
- What is the evaluation metric(s)?
  - Some tasks are straightforward to measure (e.g accuracy, for mathematical reasoning). Some are much harder (e.g LLM persuasive ability - think about how you would do this).
    - The metric we measure is Casual Relevance which is from the CoIG paper. For every chain we flip 1 attribute in a sub-prompt and use an MLLM judge to test if the change shows up and if it persists to the final image. Then we give one score per chain. But then we measure the gap between the Real, Shuffled, and Substituted chains with a paired t-test. If scores are the same throughout, then it only detects the compositional lock.

## Benchmarks/Evaluation Sets

- Evaluation and comparison with existing systems.
  - What are your baselines?

### Benchmarks and Evaluation

We will evaluate our experiment using the color-attribute subset of T2I CompBench, which is the same benchmark subset used by CoIG for its original Causal Relevance evaluation. Using the same benchmark allows direct comparison with CoIG's methodology while providing prompts with clear object attribute relationships that can be tested for causal influence.

In each condition, we produce one image chain with the original CoIG architecture, like the Compositional Strategy Planner, Autoregressive Refinement Model, compositional lock, and evaluation pipeline, which is driven by the real, correctly ordered sub-prompts. In the Shuffled and Substituted conditions, the image chain is used but the text associated with each step is re-labeled according to the modified condition, so there is no generation of new images at all. By employing the original CoIG architecture rather than replacing it with other models, we can replicate the original model configuration and avoid potential confounds due to differences in implementation in the Causal Relevance study.

These conditions allow us to separate semantic faithfulness from the effects of the compositional lock. The primary evaluation metric is ColG's Causal Relevance score. Following the original evaluation procedure, we will change intermediate sub prompts and use the authors' original MLLM judging pipeline and prompts to determine whether the modified information appears and persists in the final image. Each prompt will receive a Causal Relevance score for all three conditions.

## Baselines

The original ColG evaluation setup serves as the primary baseline, represented by the Real Chain condition. This baseline contains both semantic alignment and the compositional lock, meaning it cannot determine which factor causes high Causal Relevance scores.

The Shuffled and Substituted Chains serve as controlled negative baselines. The Shuffled Chain tests whether the metric depends on the correct ordering of steps, while the Substituted Chain tests whether the metric remains high even when semantic relationships between steps are removed.

We will compare Causal Relevance scores across the three conditions using a paired statistical test with approximately 100 T2I-CompBench color attribute prompts. If scores remain similar across all conditions, this suggests that Causal Relevance primarily measures the compositional lock. If the Real Chain significantly outperforms the negative controls, this provides evidence that the metric captures genuine step by step level semantic influence, this creates a graded and falsifiable evaluation, where different outcomes reveal whether Causal Relevance is fully influenced by the compositional lock, partially sensitive to semantic influence, or successfully measuring causal faithfulness.

## Ideal Results

- What's the best case scenario/what do you hope to demonstrate?
  - What is your hypothesis? What results would prove it to be true?

### Ideal outcomes Track 1:

We predict that disabling the compositional lock leads to immediate semantic degradation in the step-by-step images whereas the locked condition exhibits virtually perfect structural stability. Thus proving the lock is an engineering confound, justifying the necessity of Track 2.

## Ideal outcomes Track 2:

Our "Anchor Metric" proves effective in identifying certain intermediate layers that exert a disproportionately large causal effect on the resulting image. Specifically, the ideal outcome is that our Anchor Metric generates a wide range of Causal Weight scores at various intermediate steps despite the fact that ColG's original Causal Relevance metric is completely flat because of the compositional lock. Thus proving the superiority of our proposed metric over the old one.

The ideal outcome of this study is that the Causal Relevance metric produces similar scores across the Real, Shuffles and Substituted conditions, despite large differences in the semantic relationship between sub prompts and generated content. Because the compositional lock is held constant across all the conditions, this result would suggest that the metric is primarily measuring the locks preservation of generated content rather than genuine causal faithfulness between intermediate steps and the final image. Demonstrating this issue would provide an important methodological finding for the evaluation of step by step image generation systems and show the need for faithfulness metrics that can separate true causal influence from architectural constraints.

## Potential Limitations

- Computation limits, generalization limits, dataset limits, ethical limits?
- One limitation of this study is that the evaluation of Causal Relevance depends on an MLLM judge as a measure of whether visual attributes appear and persist across generation steps. Because this judge has not been independently validated against human assessments of causal relevance, inaccuracies or biases in the evaluation process could affect the interpretation of the results. A possible way to address this issue would be to compare results from multiple MLLM judges on a subset of examples to see whether they produce consistent evaluations.
- Another limitation is the challenge of fully reproducing the original ColG environment. Although we have access to the authors codebase, including the CSP, ARM, compositional lock, and evaluation pipeline, differences in model availability, API changes, or software version change since publication may require certain substitutions. Any changes from the original setup should be documented and considered when analyzing the results.
- Along with that this study evaluates Causal Relevance only within ColG's specific architecture, where the compositional lock constrains later generation steps from modifying earlier content and the findings may not generalize to other forms of visual chain of thought generation, such as diffusion based processes or unconstrained autoregressive editing, where the relationship between intermediate steps and final outputs may differ.

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## Proposal Summary: Auditing CoIG's Causal Relevance Metric

RQ: Is CoIG's compositional lock causing mechanical visual persistence, and how can we develop a better measure of true causal faithfulness during the process of generating images step by step?

The Problem: The Causal Relevance metric of CoIG might just be identifying the effect of the mechanical enforcement of the lock itself, and not semantic faithfulness. If the metric itself is completely confounded by the architecture, there is currently no way for the field to audit dynamic image verifiers.

Method (Two-Track Approach):

- Track 1 (Lock Check): A lightweight ablation experiment to check the function of CoIG's compositional lock. The lock will be temporarily turned off to establish a baseline visual drift compared to persistence enforced by the lock.
- Track 2 (Visual Faithfulness Metric): Inspired by research on "Thought Anchors" in NLP, we suggest developing a new metric of visual faithfulness. Instead of testing textual sub-prompts on a locked architecture, the proposed method would intervene in intermediate visual representations (cross-attention maps/latents) and measure the downstream visual changes to find causally determining generation steps.

Contribution: Identification of the first empirical limits of CoIG's architectural confounds and development of a new architecture-agnostic faithfulness metric for a chain of image generation.

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### Summary of PI Feedback

Overall take: The proposal has a structural weakness — if the experimental results come back as "the metric works fine", the paper has very little to contribute.

PI's recommended pivot — split effort into two parallel workstreams:

1. Lock mechanism check (small allocation of time/people). Directly verify whether the compositional lock is doing what we think it's doing. This should be relatively quick to test. Depending on what we find:

- If the lock confound is real → we have a finding
  - If not → drop it, or write it up as a small contribution.
2. Better faithfulness metric for chain-of-image generation (larger allocation). Rather than only auditing an existing metric, we can propose a new one — a faithfulness metric designed specifically for step-by-step image generation.

Suggested inspiration: Thought Anchors: Which LLM Reasoning Steps Matter? (<https://arxiv.org/pdf/2506.19143>). This paper looks at which reasoning steps in an LLM chain actually matter for the final answer.

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## Inconsistency in the Proposal

There's a critical inconsistency between the Motivation and Methods sections that we need to fix before anything else moves forward. In Motivation, the proposal says we're building unfaithful chains where "the sub-prompt's meaning does not match the content generated at that step" — which reads as keeping the *original* per-step images and only swapping the sub-prompt text on top of them. But in Methods step 4 and the diagram, the description is of *regenerating* new per-step images from the shuffled and substituted sub-prompts before running the perturbation protocol. These are two completely different experiments, and only one of them works. If we regenerate, each new sub-prompt matches its own generated step again, which means we've built a faithful chain of a different image, not an unfaithful chain — there's no ground-truth unfaithfulness left to measure, and a high Causal Relevance score would tell us nothing. The correct design is *don't regenerate*: keep the original per-step images and only swap the sub-prompt text. That gives us genuine unfaithfulness by construction, especially in the Substituted condition. One related point worth thinking about: under this corrected design, Shuffled mainly tests order sensitivity while Substituted tests full semantic mismatch — those are different things, and the proposal needs to be explicit about what each condition is measuring and why we're including both.

## Proposal Re-structuring

The current proposal is heavily reliant on finding that Causal Relevance is confounded by the compositional lock — if the results come back showing the metric actually works, we'd have very little to write about, and that's not a safe bet for a whole project.

Two parallel tracks:

- Track 1 is a lighter-weight lock mechanism check: directly verify what the compositional lock is actually doing before running the full experiment. This should be relatively quick and tells us whether the audit is worth pursuing at all.
- Track 2 is designing a new faithfulness metric for chain-of-image generation, rather than only auditing an existing one. This is valuable regardless of how Track 1 turns out and

it's a much stronger contribution overall.

To get started on Track 2, we need to read “Thought Anchors: Which LLM Reasoning Steps Matter?” (<https://arxiv.org/pdf/2506.19143>) — it studies which reasoning steps in an LLM chain actually matter for the final answer, and the intuitions there are a good jumping-off point for defining “mattering” in the image-generation setting.

### Next steps:

- Fix Methods step 4 and the diagram so it's clear the per-step images are not regenerated — only the sub-prompt text is swapped. [CHAYAN, PRANAV]
  - Rewrite the Shuffled vs. Substituted descriptions to be explicit about what each condition tests (order sensitivity vs. full semantic mismatch).
- Restructure the proposal into two tracks: a lighter Track 1 for the lock mechanism check, and a larger Track 2 for designing a new faithfulness metric. [GRACE, AKHIL, ANE]

### New faithfulness metric ideas:

#### Idea 1:

<https://arxiv.org/pdf/2506.19143>

Designing a metric that identifies/measures visual anchors

Visual anchor: specific steps or image regions that have high influence on the final output. Modifying/removing an anchor causes big shift in the final image

We can measure the impact of specific steps by ablating them during the CoIG process, and then calculating the distance between the og image and the ablated image, high delta indicates strong visual anchor

#### Idea 2:

Attention- mechanism that allows a model to look at specific parts of its input, like an internal spotlight

Attend-and-Excite: Attention-Based Semantic Guidance for Text-to-Image Diffusion Models

<https://arxiv.org/abs/2301.13826>

Summary of paper:

Each step, the model uses a mechanism called “cross-attention,” acting as a bridge between text and the image.

Authors identified that advanced generative models often fail to faithfully represent the input prompt due to catastrophic neglect, where the model fails to generate some subjects explicitly requested in the prompt, and incorrect attribute binding, where the model fails to correctly assign attributes to subjects.

Solution: the authors introduce a method called Generative Semantic Nursing (GSN), implemented as Attend and Excite

Attend and Excite: a technique that corrects the focus of a text-to-image diffusion model during the generation process.

- Every step of the denoising process, the algorithm inspects the attention maps to verify that every subject in the prompt is receiving sufficient attention.
- “Excite”: If it notices a subject is being neglected, it applies a mathematical nudge and shifts the latent code at the exact moment. By exciting the activations of the neglected token, it forces the model to allocate more of its generative budget to that specific subject.

Proposed metric: instead of exciting neglected tokens to fix an image, we use the same inspection mechanism to measure if the model is already faithfully attending to all the sub-prompts in CoIG chains. The metric measures the Spatial-Semantic Alignment between a sub-prompt’s cross-attention maps and the final generated image. A model is considered faithful if its attention spotlight at each step directly correlates with the objects in the final image.

We already have access to the CoIG codebase and the ARM generation pipeline.

- During denoising of each step, we hook into the cross-attention layers of the diffusion model to save the attention maps for each subject token in the sub-prompt

- We use a pre-trained segmentation model to generate a mask of where the object appears in the final image

- We calculate the Pearson Correlation between the attention map and the spatial mask.  
(measure of how strongly two things move together, so in this case it would be how well the heat in the model’s attention map matches the shape in the mask)

- A chain with high faithfulness will show high correlation across all the steps.

Maybe a relevant paper: <https://www.mdpi.com/2078-2489/17/2/149>

Idea 1:

Paper: <https://arxiv.org/abs/2402.16434>

Summary of Paper: The authors analyze how different stages of the denoising process contribute to the final image. They prove mathematically that not all timesteps are created equal: early to mid steps dictate the core semantic layout and global structures of the image, while later steps only refine high frequency details.

Proposed metric: Instead of using step importance to speed up the model, we use this branching mechanism as a diagnostic metric to identify exactly where a model commits to a visual plan. This metric measures how sensitive the final image is to micro changes introduced at a specific timestep. A timestep is a Visual Thought Anchor if a tiny intervention causes a massive structural divergence in the final image.

Idea 2: Similar to Grace’s second idea, more relevant papers: [arXiv:2210.04885](https://arxiv.org/abs/2210.04885),  
[WACV 2024 / arXiv:2312.01132](https://arxiv.org/abs/2312.01132) -> project page <https://shape-guided-diffusion.github.io/>

## Proposed Metric: Spatial-Semantic Alignment

To bypass the compositional lock, we must measure the model's internal reasoning trace rather than its final visual output. The Spatial-Semantic Alignment metric quantifies whether the model's internal attention exactly correlates with the spatial introduction of new pixels at a specific step.

This is calculated in three phases:

### Phase A: Computing the Delta Mask

Instead of looking at the final image, we isolate the exact spatial footprint of the work done during the current step.

Using a pre-trained segmentation model, we generate a binary spatial mask for the target attribute on the image output of the current step, and the image output of the previous step.

We calculate the Delta Mask by subtracting the previous step's mask from the current step's mask.

[Delta Mask = Current Mask minus Previous Mask]

This isolated mask contains only the newly added pixels. If the attribute was already locked in from a previous step, the Delta Mask will be empty, correctly yielding a baseline score of zero for the current step.

### Phase B: Extracting the Cross-Attention Maps

Diffusion models use cross-attention layers to bind textual tokens (the prompt) to spatial regions (the image). We hook into these layers to observe where the model is "looking" when it processes the attribute's specific text token.

Targeting the Denoising Window: Cross-attention behaves dynamically across the diffusion process. During the early steps (roughly 0-15 of a 50-step schedule), attention maps define structural layout. In later steps, attention scatters to resolve high-frequency textures.

We aggregate the cross-attention maps for the target token exclusively during this early structural window to capture the model's spatial intent, creating a composite attention map.

### Phase C: Calculating the Alignment Score

To determine if the model was faithful to the step-by-step instruction, we measure the spatial overlap between the model's internal attention and the physical change in the image (the Delta Mask).

Thresholding: Because attention maps are continuous and segmentations are binary, we binarize the composite attention map by isolating the top activations (for example, applying a threshold to keep only the top 20% most activated pixels).

Intersection over Union (IoU): We calculate the alignment using the IoU between the thresholded attention map and the Delta Mask.

[IoU = (Overlap of Thresholded Attention and Delta Mask) / (Total Area of Both Combined)]

### 3. Methodological Advantages

This metric completely eliminates the compositional lock as a confounding variable. If a model generates an attribute too early (e.g., at step 1 instead of step 2), the Delta Mask for step 2 will be empty, and the IoU score for step 2 will correctly be 0.

A high Spatial-Semantic Alignment score serves as mathematically grounded proof that the model actively dedicated its generative budget to the correct attribute at the exact step it was instructed to do so.